SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – TECHINICAL PRESENTATION

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	Sree B	**Reg. No.:**	192524023
Section / Batch:	AI and DS-2025	**Date:**	23/07/2026

Instructions to Students

- Prepare a technical presentation covering **both assigned topics**.
- Organize the content in a **clear and logical sequence** with appropriate headings and subheadings.
- Include **well-labeled diagrams, flowcharts, or concept maps** wherever applicable to explain technical concepts.
- Highlight **key concepts, formulas, algorithms, and their relationships** using suitable visuals.
- Compare related concepts and **justify the differences with appropriate technical explanations**.
- Ensure the presentation is **well-structured, visually appealing, readable, and technically accurate**.

Question Type	Focus Area	Questions	Marks
TECHINICAL PRESENTATION	*Analyze the time complexity of divide-and-conquer algorithms using the Master Theorem, with special emphasis on recurrences having logarithmic work functions such as* $T(n) = aT(n/b) + \log n$.	Q1	Q1 –100
		GRAND TOTAL:	**100 Marks**

Code of Conduct

I Sree B(192524022)_____________ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Sree B________________________

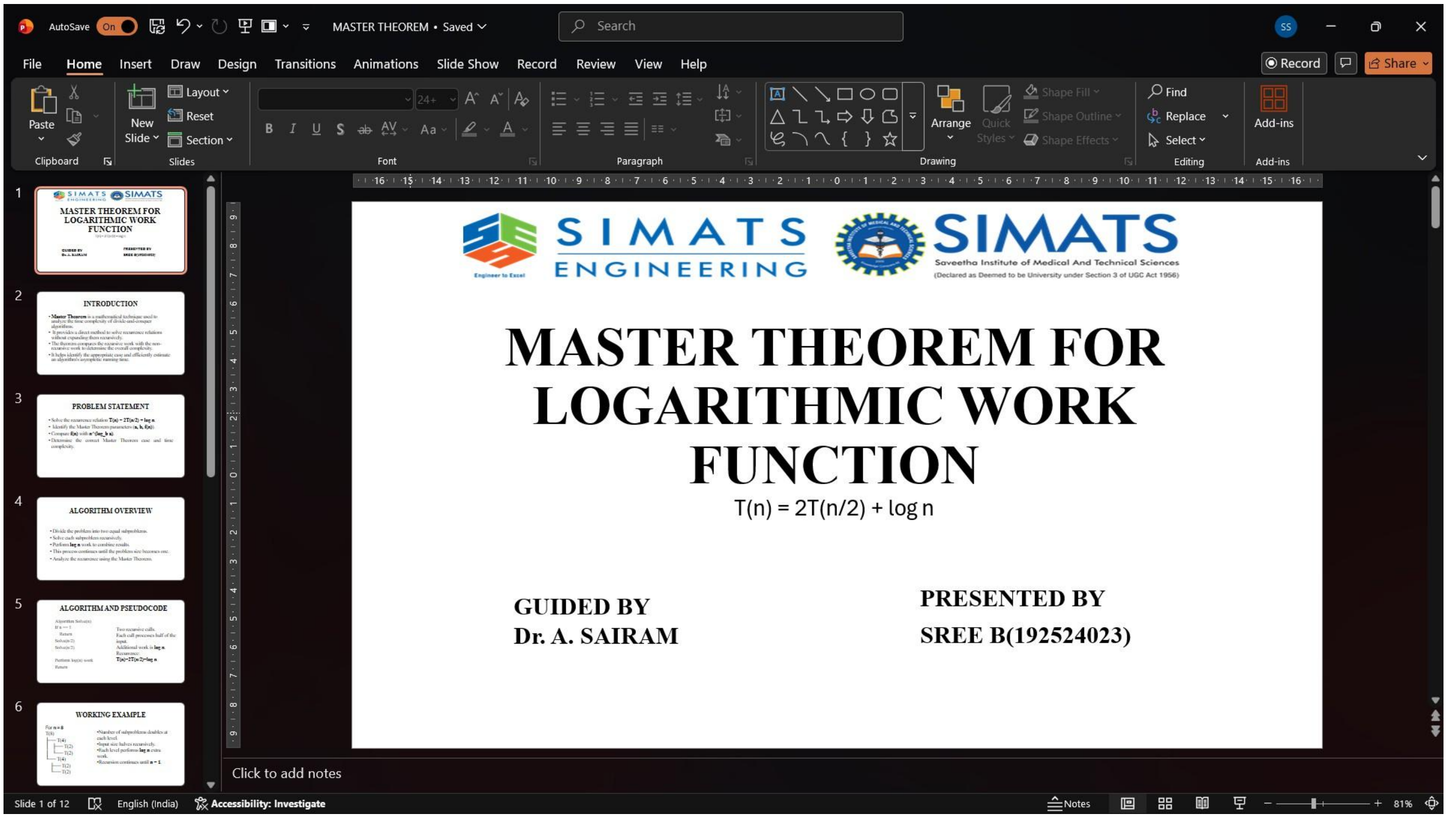
RUBRICS FOR CALCULATION

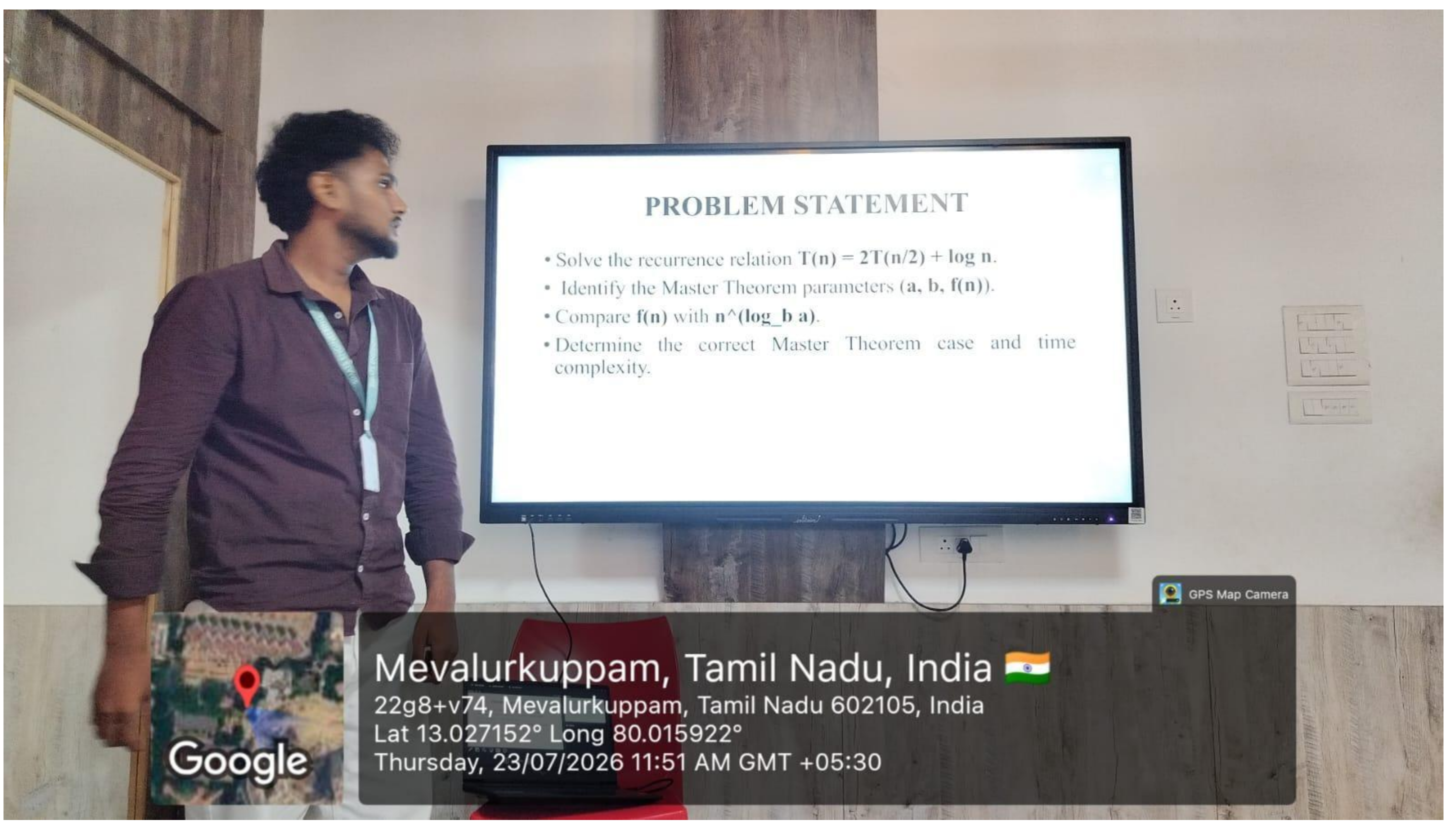
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________


MASTER THEOREM • Saved
SIMATS ENGINEERING
Engineer to Excel
SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)
MASTER THEOREM FOR LOGARITHMIC WORK FUNCTION
T(n) = 2T(n/2) + log n
GUIDED BY
Dr. A. SAIRAM
PRESENTED BY
SREE B(192524023)
INTRODUCTION
PROBLEM STATEMENT
ALGORITHM OVERVIEW
ALGORITHM AND PSEUDOCODE
WORKING EXAMPLE
Click to add notes
Slide 1 of 12
English (India)
Accessibility: Investigate


PROBLEM STATEMENT
• Solve the recurrence relation T(n) = 2T(n/2) + log n.
• Identify the Master Theorem parameters (a, b, f(n)).
• Compare f(n) with n^(log_b a).
• Determine the correct Master Theorem case and time complexity.
GPS Map Camera
Google
Mevalurkuppam, Tamil Nadu, India
22g8+v74, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.027152° Long 80.015922°
Thursday, 23/07/2026 11:51 AM GMT +05:30